In the case of large training data, a loss function $f : \mathbb{R}^n \to \mathbb{R}$ be of the form

$$f(x) = \frac{1}{N} \sum_{i=1}^{N} f_i(x), \qquad x \in \mathbb{R}^n. \tag{3.1}$$

Computing a full gradient

$$\nabla f(x) = \frac{1}{N} \sum_{i=1}^{N} \nabla f_i(x)$$

requires the evaluation of N individual gradients $\nabla f_i(x)$, which can be too expensive in case of using the gradient descent method. Instead, one could use the *stochastic gradient method* for minimizing $f : \mathbb{R}^n \to \mathbb{R}$ be: For given $x^0 \in \mathbb{R}^n$, iterate

(i) pick a random sample $d^k := d(x^k, \xi_k)$, where $\xi_k \sim \xi$ is distributed independently and identically. More precisely, $\xi_k = i(k)$ is distributed uniformly on $\{1, \dots, N\}$ and $d(x^k, \xi_k) = \nabla f_{i(k)}(x^k)$;

(ii) pick a step size $t_k > 0$;

(iii) set

$$x^{k+1} = x^k - t_k d^k.$$

The goal of this sheet is to show convergence results for stochastic gradient algoritm for smooth and convex functions under following assumptions:

(A1) $\mathbb{E}_\xi[d(x, \xi)] = \nabla f(x)$ for all $x \in \mathbb{R}^n$;

(A2) $\mathbb{E}_\xi\left[\|d(x, \xi)\|^2\right] \leq M^2$ for some $M > 0$ and all $x \in \mathbb{R}^n$.

Convergence is observed through the weighted running/ergodic mean

$$\bar{x}_\gamma^k := \sum_{j=0}^{k} \gamma_j x^j, \qquad \gamma_1, \dots, \gamma_k > 0$$

Exercise 10.1

Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth and convex and let x^* be a minimizer of f. Define $\forall k \in \mathbb{N}$, $\gamma_j = \frac{t_j}{\sum_{i=0}^{k} t_i}$

1. Show that

$$\|x^{k+1} - x^*\|^2 = \|x^k - x^*\|^2 - 2t_k \langle d^k, x^k - x^* \rangle + t_k^2 \|d^k\|^2.$$

2. Show that
$$\mathbb{E}\big[t_k\langle d^k, x^k - x^*\rangle\big] = t_k\,\mathbb{E}\big[\langle \nabla f(x^k), x^k - x^*\rangle\big]\,.$$

3. Show that
$$\mathbb{E}\big[t_k^2\|d^k\|^2\big] \leq t_k^2 M^2.$$

4. Show that
$$t_k\,\mathbb{E}\big[f(x^k) - f(x^*)\big] \leq \frac{1}{2}\mathbb{E}\big[\|x^k - x^*\|^2\big] - \frac{1}{2}\mathbb{E}\big[\|x^{k+1} - x^*\|^2\big] + \frac{1}{2}t_k^2 M^2.$$

   Hint: Use convexity of f, i.e.,
$$f(x^k) - f(x^*) \leq \langle \nabla f(x^k), x^k - x^*\rangle,$$

   together with the previous two estimates.

5. Show that
$$\mathbb{E}\big[f(\bar{x}_\gamma^k) - f(x^*)\big] \leq \frac{\|x^0 - x^*\|^2 + M^2\sum_{j=0}^k t_j^2}{2\sum_{j=0}^k t_j}.$$

   Hint: Sum the previous inequality from $j = 0$ to k and use the definition of the weighted average iterate $\bar{x}_\gamma^k$.

Exercise 10.2

Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth and convex, and let $x^* \in \mathbb{R}^n$ be a minimizer of f. Let the stepsize
$$t_k = \frac{\|x^0 - x^*\|M^{-1}}{\sqrt{k+1}},$$
Show that
$$\mathbb{E}\big[f(\bar{x}_\gamma^k) - f(x^*)\big] \leq M\|x^0 - x^*\|\frac{2 + \log(k)}{2\sqrt{k+1}}.$$

Exercise 10.3

Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth and μ-convex, and let $x^* \in \mathbb{R}^n$ be a minimizer of f. Let the stepsize
$$t_k = \frac{1}{\mu(k+1)}$$
For the empirical mean $\bar{x}^k = \frac{1}{k+1}\sum_{j=0}^k x^j$, show that
$$\mathbb{E}\big[f(\bar{x}^k) - f(x^*)\big] \leq M^2\frac{1 + \log(k)}{2\mu(k+1)}.$$

Hint: Follow 10.1 and Use strong convexity gradient inequality of f.